

Me vs ME

Complete Product & Technical Blueprint — V1

Target users: 18–30, students and early career professionals

North star: After 30 days, the user says — I understand myself better. I know my patterns.

The Problem This App Solves

Most people have felt it. You wake up with intention. By night you feel like the day slipped away. Not because you are lazy. Not because you lack willpower. Because nobody has ever shown you a clear picture of yourself.

You do not know what conditions make you sharp. You do not know which habits silently destroy your focus. You do not know what your best version of a day actually looks like — in numbers, not in vague feelings.

Me vs ME exists to answer three questions that no app currently answers together:

- What does your best day look like — exactly? What time did you wake up, what did you do, how did you feel?
- What does your average day look like, honestly? Not aspirationally. Actually.
- What is the gap — and what specifically closes it?

This is not a habit tracker. Habit trackers tell you what you did. Me vs ME tells you what it means. The entire system is designed around one insight: you already have the data inside your daily life. You just cannot see the pattern. We make the pattern visible.

North star metric: After 30 days, the user says 'I understand myself better — I know what kind of day makes me feel like my best self and what pulls me away from it.'

Section 1 — Onboarding: What We Collect and Why

Onboarding has one job: understand who this person is and what quality of life means to them — in their own words. The user defines their own north star. We track against it. Every field collected has a purpose. If it cannot be justified, it is not collected.

The moment onboarding ends, the user sees their first insight. Not after 30 days of logging. Immediately. This is the first signal that the app is different from everything else they have tried.

1.1 Core Identity — Table: users

Column	Type	Example	Why we collect it
user_id	UUID PK	uuid-v4	Primary key. Every table joins here.

Column	Type	Example	Why we collect it
full_name	VARCHAR(100)	Priya Sharma	Personalises every notification and report. 'Good morning Priya' vs 'Good morning User'.
age	INT	21	Routes to student mode (≤ 22) or career mode (≥ 23). Changes domain labels, question phrasing, example contexts.
gender	ENUM	female/male/other/skip	Optional. Used only in anonymised cohort analytics for B2B. Never shown to the user or used in scoring.
life_stage	ENUM	student / working / freelancer / homemaker / other	Single most important routing field. Decides which domains show, which questions are asked, which weekly report template is used.
sub_role	VARCHAR(100)	NEET aspirant / SDE / teacher	Free text within life_stage. Haiku reads this to personalise goal decomposition prompts and article recommendations.
city	VARCHAR(80)	Bengaluru	Timezone routing. Alarm triggers and Sunday report delivery use local time.
timezone	VARCHAR(50)	Asia/Kolkata	Auto-derived from city or device. All timestamps stored UTC, displayed local.
platform	ENUM	ios / android	Critical for feature availability. Screen time tracking and app interception available only on Android. iOS uses proxy signals.
subscription	ENUM	free / pro	Gates premium features: app blocking (V2), Personal Operating Manual (day 90), AI companion character (V2).
onboarding_complete	BOOLEAN	true	Partial onboarding = different home screen. We show what we have, not an empty state.

1.2 The Quality of Life Definition — User-Defined

This is the most important design decision in the product. Quality of life means something different for a NEET student, an SDE, and a new mother. We do not define it for them. We ask them to define it.

During onboarding, the user is shown 5 life pillars. They toggle which ones matter to them and assign relative importance. The app tracks only what they said matters. Their Me Score is built only from their chosen pillars.

Pillar	What it tracks	Who typically cares	Default weight if selected
Sleep & Energy	Sleep hours, wake quality, energy arc through the day	Everyone — health is foundational	30%
Focus & Progress	Deep work hours, goal completion, session quality	Students, professionals, creators	35%
Intentional Screen Time	Planned vs actual app usage, mindless vs intentional ratio	Anyone who feels phone is the problem	20%
Movement & Body	Steps, active minutes, physical activity	Fitness-focused users, those who sit all day	15%
Real-world Time	Phone-free time, offline activity labels	Users who want more presence outside screens	Optional add-on

User adjusts the sliders. Sum must equal 100%. This is stored in `user_domains` with `weight_in_score`. Their Me Score formula is literally built from their answer here.

Design rule: We never show a pillar the user said does not matter to them. An irrelevant metric in the score destroys trust. If a user says movement does not matter, steps never appear in their report.

1.3 Baseline Calibration — Table: onboarding_baseline

Five minutes of questions. Result: a synthetic baseline before any real data exists. User sees their first Me Score the moment they finish onboarding. The gap between current and best self is visible immediately.

Column	How it is asked	Why it matters
<code>best_month_text</code>	Think of a time in the last year when you felt really on top of things — even for 2 weeks. What was different?	Claude Haiku reads this and extracts: domains mentioned, conditions described, emotional tone. Builds the personal best anchor immediately.
<code>best_focus_hours</code>	On your best study/work day, how many truly focused hours did you get? (slider 1–12)	The personal best ceiling for focus domain. Gap between this and today = first insight.
<code>current_focus_hours</code>	Right now on an average day, how many focused hours do you actually get? (slider)	Combined with best to compute initial focus gap score shown at end of onboarding.
<code>best_sleep_h</code>	During that best period, how many hours were you sleeping? (slider 4–10)	Personal best sleep baseline. Health score cannot have a reference without this.
<code>current_sleep_h</code>	Right now, how many hours do you sleep most nights? (slider)	Current sleep estimate before any Health API data.
<code>wake_feeling</code>	When you wake up, how do you usually feel? (Exhausted / Groggy / Okay / Energised)	Sleep quality proxy. Maps to 1–4. Used in health index until Health API arrives.

Column	How it is asked	Why it matters
primary_challenge	What gets in your way most? (Phone / Low energy / Inconsistency / No motivation / Cannot focus)	Routes first week notifications and article suggestions. Phone → screen insights first. Energy → sleep insights first.
quality_of_life_pillars	JSONB: {sleep: 0.30, focus: 0.35, screen: 0.20, movement: 0.15}	The user-defined weights. This is their constitution. The entire scoring system is built from this.
synthetic_me_score	FLOAT — computed immediately from above	First Me Score. Shown at end of onboarding as: 'You are at 52% of your best self right now. That means 48% is still available.'

Section 2 — Full Daily User Journey

Every interaction from waking up to sleeping. Every screen. Every data point. Every table it goes into. This is the full picture of what a user day looks like inside the app.

2.1 Wake Up — Alarm and Snooze Tracking

The user sets an alarm the night before inside the app. The morning begins here. This is the first data point of the day and it sets context for everything that follows.

What happens: alarm rings → user snoozes 0–N times → user taps 'I am awake' or types their wake time → snooze count and actual wake time are recorded.

Table: alarm_events

Column	Type	Example	Why important
alarm_id	UUID PK	uuid	Unique alarm instance.
user_id	UUID FK	uuid	Links to user.
alarm_type	ENUM	wake / sleep	Wake = morning. Sleep = bedtime reminder. Same table, two purposes.
scheduled_time	TIMESTAMPTZ	06:00:00	What time user planned to wake up.
actual_wake_time	TIMESTAMPTZ	06:15:00	When user tapped 'I am awake'. The real wake time.
snooze_count	INT	3	Times snoozed before waking. Insight: 'You average 3.1 snoozes before waking — article on why each snooze makes you more tired.'
snooze_total_minutes	INT	15	Total extra time in snooze cycles. Used in sleep debt calculation.
date	DATE	2025-01-15	For daily aggregation and weekly report.

What this builds: wake time consistency (variance across 7 days), snooze pattern insight, correlation between snooze_count and session quality later that day.

2.2 Morning Phone Behaviour — Screen Timer

After waking, the highest-risk behaviour is immediately opening Instagram, YouTube, or WhatsApp. This sets the mental tone for the entire morning. We track it and gently intervene.

V1 implementation: after 'I am awake' tap, a 30-minute awareness window begins. Android uses UsageStatsManager to track which apps are opened. iOS users see a self-report prompt. No blocking in V1 — pure awareness and tracking.

Table: morning_screen_events

Column	Type	Why important
phone_free_minutes	INT	Minutes after waking before first guarded app was opened. Core signal.
first_app_opened	VARCHAR(100)	First app opened after alarm. Most predictive signal for morning context.
distracting_apps_opened	TEXT[]	List of guarded apps opened in first 30 min. e.g. ['instagram','youtube']
total_screen_min_first_30	INT	Total phone usage in first 30 minutes. Weekly average shown in report.
date	DATE	For daily aggregation.

2.3 Morning Stamp — Intention Setting

Around 8–9 AM. Twenty seconds. Sets the context for the entire day. The app asks: what kind of day is today, and how are you going in?

Table: morning_stamps

Column	Type	Example	Why important
day_type	ENUM	deep_focus / mixed / recovery	User's intention for today. Recovery days scored differently — we do not penalise planned rest.
energy_level	INT 1–5	4	Subjective energy at start of day. Key predictor when no wearable exists.
mood	INT 1–5	3	Optional emotional state. Correlates with focus quality and intentional screen use.
goal_for_today	TEXT	Complete trees chapter + 2 LeetCode	Evening check-in references this specifically: did you do what you said you would?
stamp_time	TIMESTAMPTZ	08:42:00	Late stamps (after 10 AM) correlate with lower intent accuracy. Pattern surfaced in weekly report.

2.4 Intentional App Use — The Popup Intervention

When the user opens a guarded app (Instagram, YouTube, Reels, etc.), a popup intercepts before the app opens. This is V1 medium intervention: awareness, not blocking.

The popup shows: 'Hey — why are you opening this?' with a dropdown. The user picks a reason and sets a time limit. A notification fires when the limit is reached. End of day, the data feeds into the daily report.

Table: app_open_intents

Column	Type	Example	Why important
app_package	VARCHAR	com.instagram.android	Which app was intercepted.
reason	ENUM	entertainment / boredom / specific_task / habit / stress_relief / other	The declared reason. Boredom and stress_relief are the two most correlated with later regret. We do not say this — we show the pattern.
intended_minutes	INT	30	How long they said they would use it.
actual_minutes	INT	67	How long they actually used it (from UsageStats). The gap is the data.
time_of_day_hour	INT	22	Hour of the day (0–23). Pattern: boredom scrolling clusters at 10–11 PM for most users. Surfaced after 21 days.
mindless_flag	BOOLEAN	true	Computed: if reason is boredom/habit/stress_relief AND actual > intended × 1.5, flag as mindless. Only mindless usage counts against screen score.
opened_at	TIMESTAMPTZ	2025-01-15 22:04:00	Timestamp for time-of-day pattern analysis.

What this builds weekly: 'You opened Instagram 34 times this week — 18 for entertainment, 12 out of boredom, 4 for a specific reason. Mindless usage ratio: 71%. Your boredom scrolling happens mostly between 10–11 PM, right before sleep.'

V2 premium: actual app blocking after the time limit. Android: AccessibilityService overlay. iOS: Screen Time API (limited) or VPN profile (more reliable). This ships as paid feature.

2.5 Focus Sessions — Deep Work Tracking

The core career/learning data. User taps Start when beginning focused work, taps End when finished. One session = one continuous block. Sessions are tied to goals and subtasks.

Table: focus_sessions

Column	Type	Example	Why important
goal_id	UUID FK nullable	uuid	Which goal this session contributes to. Null = general focused work.

Column	Type	Example	Why important
subtask_id	UUID FK nullable	uuid	Which subtask specifically. Enables per-subtask completion tracking.
planned_duration_min	INT	60	How long user intended before starting. Set at session start.
actual_duration_min	INT	72	Computed: end_time – start_time in minutes.
quality_rating	INT 1–5	4	Self-rated quality immediately at end. 1=scattered, 3=solid, 5=flow state.
focus_score	FLOAT	288	Computed: actual_duration_min × quality_rating. A 72min quality-4 session = 288. A 180min quality-1 session = 180. Quality wins.
execution_ratio	FLOAT	0.83	min(planned, actual) / max(planned, actual). 60 planned, 72 actual = 0.83. Tracks planning accuracy over time.
context_location	ENUM nullable	home / library / cafe	Optional. Builds environmental performance data. After 30 days: 'Your library sessions average quality 4.1 vs home sessions 2.9.'

Focus score formula justified: Ericsson & Krampe (1993) deliberate practice research shows quality of engagement, not raw time, predicts performance growth. A 1-hour quality-5 session (score: 300) outperforms a 2-hour quality-2 session (score: 240). The formula rewards real focus, not clock time.

2.6 Goal and Subtask System

User creates a goal in natural language. Claude Haiku decomposes it into milestones and subtasks. The user edits freely. Progress ticks come from notifications — no need to open the app.

Table: goals

Column	Type	Why important
title	TEXT	Original text the user typed. Preserved exactly. 'Crack NEET 2025 with 650+'
deadline	DATE	Target completion. Used for velocity calculation: are they on track?
is_daily	BOOLEAN	Daily goals reset every day. Scored differently — continuity matters, not completion of a fixed scope.
custom_days	INT nullable	For 2-day or custom-duration goals. Null for ongoing.
sticky_notification	BOOLEAN	Persistent progress bar in notification tray. User opts in. Keeps goal visible without opening app.
total_progress_pct	FLOAT	Computed from subtask weights × completion. Updated on every subtask tick.

Column	Type	Why important
status	ENUM	active / paused / completed / abandoned. Abandoned goals tracked separately — pattern insight if user consistently abandons at same stage.

Table: subtasks

Column	Type	Why important
milestone	TEXT	Milestone group. e.g. 'Foundation Phase'. Groups related subtasks.
title	TEXT	The subtask name. AI-generated but user-editable.
est_days	INT	AI-estimated days to complete. Used to compute weight.
weight	FLOAT	$\text{est_days} / \text{sum}(\text{all subtask est_days})$. A 45-day subtask in a 180-day goal = weight 0.25 = 25% of total progress.
progress_pct	INT 0–100	User-updated via notification tick or in-app. This is the primary daily input for goal progress.
time_blocks	JSONB	e.g. <code>[{start:'09:00',end:'11:00'}]</code> . Time blocks user assigned. Notification fires at start of each block.

Progress calculation: $\text{goal_progress} = \text{sum}(\text{subtask.weight} \times \text{subtask.progress_pct} / 100)$.
 Example: 4 subtasks at 25% weight each. Subtask 1 = 100%, Subtask 2 = 60%, rest = 0%.
 Goal = $(0.25 \times 1.0) + (0.25 \times 0.6) = 40\%$.

Velocity: $\text{expected_pct} = (\text{days_elapsed} / \text{total_days}) \times 100$. Velocity gap = actual – expected.
 +10% = ahead. –10% = behind with projection: 'At current pace you finish 11 days after deadline.'

2.7 Midday Energy Check — 1:00 PM

Optional push notification. One tap. Builds the intraday energy curve: morning → midday → evening. Three data points per day, averaged over a week, reveal the user's personal energy pattern.

After 21 days: 'Your energy peaks at 9 AM and dips at 1–2 PM before partially recovering. Schedule deep work before noon. Administrative and low-cognitive tasks for the afternoon.'

Table: midday_energy_checks

Column	Type	Why important
energy_level	INT 1–5	Middle point of daily energy arc. Combined with morning and evening to reveal personal energy pattern.
ts	TIMESTAMPTZ	Actual time of tap. Some users do this at 12:30, others 2 PM. We use actual time, not scheduled.
date	DATE	For daily aggregation.

2.8 Daily End-of-Day Report — The Mirror

This is the most important screen in the app. It is not a check-in. It is a mirror. The app shows the user what their day actually looked like — in data — and asks them how it felt.

Shown at ~9 PM. The user sees a summary card: when they woke up, how many snoozes, morning phone-free time, focus hours, goal progress, phone usage vs intention, phone-free time and what they labelled it.

At the bottom: two sliders. These are the daily subjective inputs that give meaning to all the objective data.

The Productiveness and Contentment Scale — Defined

We show two ratings, not one. They measure different things and their divergence is data.

Rating	Question asked	What 1 means	What 3 means	What 5 means
Productiveness	How productive did you feel today? (1–5)	I barely got anything done. The day went by and I have nothing to show for it.	I did some things. Not my best, not my worst. Average.	I was locked in. I finished what I set out to do and felt the flow.
Contentment	How content are you with how you spent your day? (1–5)	I feel like I wasted today. I wish I had done things differently.	It was okay. I used my time reasonably, even if not perfectly.	I am satisfied. The way I spent today aligns with what matters to me.

Why two separate ratings: a student can score productiveness 5 (focused 6 hours, completed all subtasks) and contentment 2 (but missed a friend's birthday and feels hollow). The divergence IS the insight. We surface this pattern: 'On your high-productiveness days, your contentment rating was 2.1 on average. On balanced days it was 3.8. Something to reflect on.'

We never use the word regret. We use contentment. The insight can be about regret. The language never is. This is a hard rule.

Table: evening_checkins

Column	Type	Why important
day_summary_shown	JSONB	The full data card shown to user: wake_time, snooze_count, focus_hours, goal_progress, screen_summary, phone_free_intervals.
productiveness_rating	INT 1–5	How productive the user felt. Most honest daily performance signal. Correlates with focus_hours and goal_progress.
contentment_rating	INT 1–5	How content with how they spent the day. Divergence from productiveness = the most interesting weekly signal.
intent_goal_completed	BOOLEAN nullable	Did they do what they wrote in morning stamp? Only asked if goal_for_today was filled.
blocker_text	TEXT nullable	What got in the way? Free text, optional. 'YouTube after lunch, about 2 hours.' Stored verbatim for pattern analysis.
energy_evening	INT 1–5	Third point of daily energy arc. Evening energy low while morning was high = energy drain identified.

Column	Type	Why important
phone_free_label	TEXT nullable	What did user do during their biggest phone-free block? 'Watching TV / playing cricket / hospital / studying'. Free text.

2.9 Phone-Free Time Tracking

When the phone is not being used, we log it. We do not assume what the user was doing. At end of day, we show them the intervals and ask them to label the biggest one. That label is their data.

Note: watching TV, being on a laptop, being on someone else's phone — user can write this. We categorise these as screen-adjacent, not phone-free. The label is theirs. We just make it visible.

Table: phone_free_intervals

Column	Type	Why important
start_time	TIMESTAMP TZ	When phone went idle (screen off + no interaction).
end_time	TIMESTAMP TZ	When phone was picked up again.
duration_minutes	INT	Computed: end – start. Only intervals ≥ 30 minutes are logged. Sleep is the largest — excluded from daytime calculation.
activity_label	TEXT nullable	User label from evening check-in. 'Playing cricket', 'Watching TV', 'Trip to hospital'. Free text, user-defined.
is_sleep	BOOLEAN	True for the overnight interval. Excluded from phone-free time report — sleep is tracked separately.

2.10 Sleep Initiation — Bedtime Tracking

Mirror of the morning alarm. User declares sleep, the screen timer begins, we see how much they actually use their phone after declaring sleep. This data feeds the weekly report, not the daily — we do not shame same-night.

Table: sleep_events

Column	Type	Why important
bedtime_target	TIMESTAMP TZ	What time user aimed to sleep.
sleep_declared_time	TIMESTAMP TZ	When user tapped 'Going to sleep'.
post_declaration_screen_min	INT	Minutes of phone usage after declaring sleep. Android: UsageStats. iOS: screen interaction proxy. Shown in weekly report only.
apps_used_after_declaration	TEXT[]	Which apps were used after 'going to sleep' tap. Weekly insight: 'You spent 22 min on YouTube after declaring sleep on 4 nights this week.'
actual_sleep_time	TIMESTAMP TZ nullable	Estimated from HealthKit/Google Fit sleep detection, or when screen went permanently off.

Column	Type	Why important
sleep_latency_min	INT	Computed: actual_sleep_time – sleep_declared_time. High latency consistently → article on sleep hygiene.

Section 3 — Passive Data Collection

Data collected without any user action. The app works even on days the user does not log. Health data comes from APIs. Screen data comes from OS-level tracking on Android.

3.1 Table: passive_health (populated nightly)

Column	Source	Why important
sleep_duration_h	Apple HealthKit HKCategoryTypeIdentifierSleepAnalysis / Google Fit TYPE_SLEEP_SEGMENT	Core health metric. Most predictive variable in the entire system for next-day performance.
sleep_quality_estimate	Derived: <5h=1, 5-6h=2, 6-7h=3, 7-8h=4, >8h=5	Quality proxy when no wearable HRV data is available. Used in health index.
steps	HealthKit HKQuantityTypeIdentifierStepCount / Google Fit TYPE_STEP_COUNT_DELTA	Physical activity proxy. 10,000 steps = full movement score. WHO 2020 guidelines.
active_minutes	HealthKit AppleExerciseTime / Fit TYPE_ACTIVITY_SEGMENT	More precise than steps. 30+ min daily = positive health signal.
resting_hr	HealthKit if wearable connected — nullable	Recovery indicator. Elevated resting HR suggests poor recovery.
hrv_ms	HealthKit if Apple Watch / Oura — nullable	Best recovery metric available. Null for most V1 users. Reserved for future.

3.2 Table: passive_screen (Android only, populated nightly)

Column	Source	Why important
app_package	UsageStatsManager.queryUsageStats()	Which app. e.g. com.google.android.youtube
actual_minutes	UsageStats.getTotalTimeInForeground()	Real usage time in minutes.
intention_minutes	From user_screen_intentions table	What user said they wanted to limit this app to. Null if no intention set.
adherence_pct	$\min(\text{intention}, \text{actual}) / \text{intention} \times 100$	The key metric. 100% = used exactly as planned. >100% = went over. Null if no intention set.

Column	Source	Why important
launch_count	UsageStats open count	High launch count + low time = compulsive checking behaviour. e.g. opened Instagram 18 times, 4 min each.

Section 4 — Daily Score Aggregation

Every night at 11:59 PM, a background job computes daily_scores from that day's raw events. This is the table all algorithms read. Not the raw event tables. The aggregation layer is what makes the system fast and explainable.

Table: daily_scores

Column	Formula	What it represents
focus_score	SUM(session.actual_duration_min × session.quality_rating) for all sessions	Weighted deep work. The career domain's primary signal.
focus_hours	SUM(session.actual_duration_min) / 60	Raw hours. Used in weekly report for readability.
execution_ratio	AVG(min(planned,actual)/max(planned,actual)) across sessions	Planning accuracy. Are estimates improving over time?
goal_progress_delta	SUM(goal progress change today across all active goals)	How much total goal progress made today. Engagement signal.
productiveness_rating	From evening_checkins	Subjective productivity 1–5. Self-truth.
contentment_rating	From evening_checkins	Subjective contentment 1–5. Life quality signal.
energy_am / midday / pm	From respective tables	3-point energy arc. Averaged = personal energy pattern over time.
sleep_h	From passive_health.sleep_duration_h for PRIOR night	Sleep the night before. Lag-1 predictor for today's performance.
health_index	(sleep_score × 0.5) + (energy_score × 0.3) + (steps_score × 0.2)	Composite health signal. 0–100 scale.
screen_mindless_ratio	mindless_app_minutes / total_app_minutes for guarded apps	Mindless ratio. Core screen score input.

Column	Formula	What it represents
phone_free_daytime_h	SUM of phone_free_intervals ≥30 min excluding sleep	Real-world time. Shown in weekly report.
snooze_count	From alarm_events	Morning discipline signal. Correlates with session quality.
morning_phone_free_min	From morning_screen_events.phone_free_minutes	Morning phone behaviour. Correlates with focus score.

Health Index Calculation

Computed daily. Each sub-score normalised to 0–100 before weighting.

```

sleep_score = min((sleep_h / 8.0) × 100, 100)           – NSF 2015: 8h = optimal
energy_score = (avg(energy_am, midday, pm) / 5) × 100   – available readings
averaged
steps_score = min((steps / 10000) × 100, 100)          – WHO 2020: 10,000 steps
health_index = (sleep_score × 0.50) + (energy_score × 0.30) + (steps_score × 0.20)

```

Sleep carries 50% weight because Walker (2017) and Van Dongen et al. (2003) demonstrate sleep is the single most powerful physical variable affecting next-day cognitive performance. Energy (30%) captures what physiology misses — the subjective felt state. Steps (20%) adds physical activity signal.

Me Score — Weekly Composite

Computed every Sunday from the week's daily_scores. This is the headline number.

```

Step 1: domain scores
focus_domain = (week_avg_focus_score / personal_best_focus_score) × 100
health_domain = week_avg_health_index
screen_domain = 100 - (week_avg_screen_mindless_ratio × 100)
goal_domain   = AVG(goal_velocity_score) across active goals

Step 2: weighted by user's own pillar weights from onboarding
me_score = Σ (domain_score × user_defined_weight)

Step 3: floor
me_score = max(20, computed_me_score)   – never below 20

Step 4: personal best
if me_score > personal_best → update. Personal best only moves up.

```

Section 5 — Analytics: Deterministic First, ML Where It Earns It

This is the most important architectural decision in the system. Most apps use ML for everything and produce black-box outputs users do not trust. Me vs ME does the opposite: every insight that can be computed with arithmetic is computed with arithmetic. ML is reserved only for patterns that arithmetic cannot find.

The result: 70% of insights are deterministic — fast, explainable, accurate on day 1. 30% are ML-derived — personal, cross-variable, and only shown once enough data exists to make them valid.

5.1 Deterministic Analytics — What Runs From Day 1

These require no training data. No minimum days. They run immediately and are always trustworthy.

Streaks — Consistency Signal

Definition: consecutive days where a given signal met or exceeded the user's own baseline.

- Focus streak: N consecutive days where `focus_hours` $\geq$ user's 7-day average
- Sleep streak: N consecutive nights where `sleep_h` $\geq$ 6.5h (floor) or user's baseline
- Phone-free morning streak: N consecutive days where `phone_free_minutes` $\geq$ 20
- Contentment streak: N consecutive days where `contentment_rating` $\geq$ 3

Shown in weekly report with a simple sentence. No gamification points, no badges. Just the number and what it means.

Rule-Based Triggers — Instant Insights

Condition (deterministic rule)	What the app shows
<code>sleep_h</code> < 6.0 for 3 consecutive days	'You have slept under 6 hours for 3 days in a row. Research consistently links this with 30% reduced cognitive performance the next day. Article on sleep debt linked below.'
<code>snooze_count</code> > 3 for 5+ days in a week	'You snoozed an average of 4 times before waking this week. Each snooze cycle fragmenting sleep is making you more tired, not less. Here is why.'
<code>morning_phone_free_min</code> < 5 for 5+ days	'Your phone was in your hands within 5 minutes of waking every day this week. This sets your brain's dopamine baseline for the rest of the morning. Article on why mornings are protected.'
<code>contentment_rating</code> < 2 for 3 consecutive days	'Three tough days in a row. Your data shows focus was actually good — the contentment gap might be worth reflecting on. What does a satisfying day look like to you?'
productiveness high + contentment low	'Your productiveness was 4.2 this week but contentment was 2.1. High output, low satisfaction. This pattern has appeared before — your data, not a judgment. Something to sit with.'
<code>goal_velocity_gap</code> < -15% for 2 weeks	'You are 15% behind on [goal name]. At current pace, you will finish 18 days after your deadline. One focused session added per week closes this gap.'

Trend Lines — Is This Week Better Than Last?

Simple linear comparison. No model. Just arithmetic.

- Weekly Me Score delta: current – previous week. '+6 from last week' or '-4 from last week'.
- 4-week focus trend: is weekly average focus_hours going up, flat, or down?
- Sleep trend: is sleep_h improving, stable, or declining across last 4 weeks?
- Contentment trend: is the weekly average contentment_rating moving?

Shown as a direction signal, not a grade. Up arrow. Flat line. Down arrow. The story underneath is the data.

5.2 ML Algorithms — Only Where Rules Cannot Reach

Five algorithms. Each one is used only when deterministic analysis cannot produce the insight. Each has a minimum data requirement before it shows anything.

Algorithm 1: EWMA — Rolling Baseline

What it solves: the user's baseline needs to update every day to reflect who they are now, not who they were 3 months ago. Simple average is too slow. Yesterday alone is too noisy.

```
Formula:  $S_t = (0.1 \times \text{today\_score}) + (0.9 \times \text{yesterday\_baseline})$   
Effective memory window: ~19 days (Hunter 1986)  
Implementation: pandas.DataFrame.ewm(span=19).mean() — one line
```

What feeds it: daily_scores.me_score, focus_score, health_index, contentment_rating, screen_mindless_ratio. One EWMA per metric per user. Updates nightly. Stored in user_baselines.

Why this matters: every other algorithm in the system uses the EWMA baseline as its reference point. CUSUM needs it. Correlation needs it. The weekly report delta is baseline-to-baseline, not raw-to-raw. Without EWMA, all comparisons are noisy.

Source: Roberts, S.W. (1959). Control Chart Tests Based on Geometric Moving Averages. Technometrics, 1(3), 239–250. Used in financial volatility (Bloomberg), manufacturing quality control, and Apple Watch resting heart rate trending.

Algorithm 2: Pearson Correlation with 1-Day Lag

What it solves: finding which conditions predict the user's performance. Not for everyone — for this specific user.

$$r = \frac{\sum [(X_i - \bar{X})(Y_i - \bar{Y})]}{\sqrt{[\sum (X_i - \bar{X})^2 \times \sum (Y_i - \bar{Y})^2]}}$$

Variable pairs tested: sleep_h[day-1] vs focus_score[day], morning_phone_free_min vs session quality, snooze_count vs contentment_rating, mindless_ratio vs productiveness_rating, energy_am vs goal_progress_delta.

Guard conditions: $N \geq 21$ data points, $|r| > 0.5$, $p < 0.05$ (Cohen 1988 power analysis). Below these thresholds, nothing is shown. The user never sees a spurious correlation.

```
Library: scipy.stats.pearsonr(x, y) — returns (r_value, p_value)
```

Output stored in user_correlations: (user_id, feature_x, feature_y, r_value, p_value, computed_at). Weekly report 'what you did not know' section reads from this table.

Source: Pearson (1895) + Granger (1969, Nobel 2003) for lag-1 causal direction. Used by Oura Ring for readiness score, Whoop for recovery prediction.

Algorithm 3: CUSUM — Drift Detection

What it solves: detecting when a user is in a sustained decline — not one bad day, but a real drift away from their baseline that a daily check would miss.

```
C-t = max(0, C-{t-1} - (Xt - μ0 + k))  
where μ0 = EWMA baseline, k = 0.5σ, alert when C-t > h = 5σ
```

Triggers: 7 consecutive days below baseline → gentle nudge in weekly report. 14 consecutive days → deeper wellbeing check-in. Language is always: 'here is what changed, here is one small thing.' Never 'you are doing badly.'

```
Library: ruptures (pip install ruptures) or manual Python — 10 lines
```

Source: Page, E.S. (1954). Continuous Inspection Schemes. Biometrika, 41(1/2), 100–115. Used in ICU early deterioration detection and network anomaly detection.

Algorithm 4: K-Means Clustering — Personal Operating Manual

What it solves: after 90 days, finding what conditions defined the user's best performance days. Not what they should do. What they actually did on their best days.

Process: take top 25% of days by me_score. Cluster their conditions. The centroid = 'your best day profile'. Shown as: 'Your best days cluster around: sleep 7–7.5h, first session before 9:30 AM, morning phone-free 25+ min. These conditions were present on 87% of your top days.'

```
Input: [sleep_h, energy_am, first_session_start_h, screen_mindless_ratio,  
snooze_count]  
K = 2 (best days cluster vs rest). N requirement: ≥ 90 days, ≥ 15 top-quartile days  
Library: sklearn.cluster.KMeans(n_clusters=2).fit(X)
```

This is the premium unlock at day 90. The Personal Operating Manual. Conditions the user did not know about themselves — surfaced from their own 3 months of data.

Source: MacQueen (1967), Lloyd (1982). Used by Spotify for listening pattern clusters, Netflix for taste profiles, Whoop for recovery pattern identification.

Algorithm 5: Thompson Sampling — Notification Timing

What it solves: finding the time each user actually responds to notifications, without asking them. The app learns which hour gets a response within 30 minutes and converges on it.

```
For each hour slot h: track αh (responses) and βh (non-responses)  
Sample θh ~ Beta(αh+1, βh+1). Send notification at argmax(θh)
```

Does not fully lock in — continues occasionally exploring other times in case schedule changes. Explore-exploit balance built in.

```
Library: scipy.stats.beta.rvs(alpha, beta) for sampling
```

Source: Thompson (1933). Used in production at LinkedIn for notification timing (LinkedIn Engineering Blog, 2018), Duolingo for streak reminder timing.

Section 6 — Weekly Report

Generated every Sunday at 8 PM. Reads from: daily_scores (7 rows), weekly_reports history, user_correlations, user_baselines, goals + subtasks, phone_free_intervals with labels. The report tells a story — not a dashboard. One scroll from top to bottom = one coherent narrative about the week.

Language rule throughout the report: this person is trying to understand themselves. Every sentence is framed as observation, not judgment. We show patterns. We do not label people.

Section 1 — Me Score Headline

Large number. Delta from last week (+6 or -4). Percentage of personal best. Personal best week shown for context.

Language: never '74 out of 100'. Always 'You are at 81% of your March baseline. Your best week was 91 — that is still the ceiling.'

Section 2 — Daily Pattern Bar Chart

7 bars, one per day. Height = daily_scores composite. Green if above 30-day EWMA baseline, grey if below. No red. No shame. Below the chart: '30-day average: 61. This week: 63. Trending up.'

Section 3 — How You Spent Your Days

Wake time consistency. Snooze average. Morning phone-free average. Phone-free daytime hours with labels — what you were doing when you were off your phone. This section shows reality without editorial comment.

Example: 'Your phone-free time this week: 14.2h across 8 intervals. Labels you gave: studying (4h), sleeping early (3h), cricket with friends (2h), watching TV (2.5h), hospital visit (2.7h).'

Section 4 — Energy Arc

Average morning / midday / evening energy across the week as a simple 3-point curve. Pattern shown if consistent: 'Your energy peaks at 9 AM, dips sharply at 1 PM, partially recovers at 7 PM. Across 4 weeks this pattern is consistent. Deep work belongs in the morning.'

Section 5 — Focus and Goal Progress

Focus hours bar chart for the week. Session quality average. Intent accuracy from morning stamp vs evening actual. Goal velocity per active goal — on track, ahead, or behind with projection.

Best session of the week always surfaced with positive framing: 'Thursday: 2.8h, quality 4. Your best session in 3 weeks. That version of you showed up this week.'

Section 6 — Screen and App Behaviour

Total mindless vs intentional app usage for the week. Per-app breakdown for guarded apps. Time-of-day pattern if 21+ days of data: 'Your boredom scrolling clusters at 10–11 PM, right before sleep — 4 nights this week.'

Post-sleep phone use shown here, not daily: 'You used your phone for an average of 19 minutes after declaring sleep this week, mostly YouTube. Your sleep latency on those nights was 28 minutes vs 11 minutes on clean nights.'

Section 7 — The Pattern You Did Not Know

One correlation-derived insight per week. The section no other app has. Not from a content library. From their own data.

Format: '[Condition X] and [outcome Y] move together in your data. When [X], your [Y] is [r×100]% higher. This pattern appeared [N] times in your last 4 weeks.'

If no correlation confirmed yet (< 21 days): population finding clearly labelled as general research, not personal. 'We do not have enough of your data for a personal pattern yet. Here is what research shows generally.'

Section 8 — The Productiveness vs Contentment Gap

This is the most personal section. If the two ratings diverge consistently, we name it.

'This week your average productiveness was 4.1. Your average contentment was 2.3. That gap of 1.8 has appeared in 3 of the last 4 weeks. High output but low satisfaction is a real pattern in your data. What does a satisfying day actually look like to you? This is worth reflecting on.'

If they converge (both high or both low): 'Productiveness and contentment moved together this week — both around 3.8. Your output and how you feel about it are aligned.'

Section 9 — The Win

Non-negotiable. Every report ends here. The best day of the week. Sourced from: MAX(me_score) for the week.

'Wednesday was your best day this week. You woke without snoozing, stayed phone-free for 40 minutes, completed 3.1h of focused work, and rated contentment 4. That is who you already are. The question is what conditions put that version of you on the field.'

If the whole week was low: 'This week was hard. You still showed up every day to log. That is not a small thing. Showing up when things are difficult is exactly how the patterns you need to see get built.'

Section 10 — One Thing for Next Week

Not a list. One. Selected algorithmically: find the domain with the largest gap vs personal best this week, find the metric that dropped most, find the simplest correlated behaviour that predicts improvement.

Format: 'One thing: [specific action]. Why: [data reason from this user's own history, or population evidence clearly labelled as such].'

Two buttons: I will try this (sets a reminder, logged in commitments table) and Skip for now. Follow-through rate tracked as a product engagement metric.

Baumeister et al. (1998) decision fatigue research: multiple recommendations reduce compliance to near zero. One clear action with a personal data reason outperforms a list of five every time.

Section 7 — Monthly Report

The weekly report shows this week. The monthly report shows the arc. What changed over 4 weeks. What the data reveals that could not be seen in a single week. Sent on the last Sunday of each month.

What the monthly has that weekly does not

- Month-over-month trend: is this month's average Me Score above or below last month? Is the trend line across 4 weeks going up, flat, or down?

- Biggest pattern of the month: the correlation that needed 30+ data points to emerge. Only surfaces here. 'This month we have enough data to say: your best focus days in October had one thing in common — you slept before midnight the night before. 9 of your 10 best focus days. That is your pattern now.'
- Personal bests of the month: best day, best week, best single session, best sleep week. Not graded. Just identified and named.
- Behaviour shift report: did anything change this month vs last month? Snooze count trending down? Morning phone-free time improving? Contentment gap closing? Shown as directional changes, not scores.

Table: monthly_reports

Column	What it stores
month_start	DATE — first day of the month. Primary key with user_id.
avg_me_score	Average weekly Me Score across the month's 4 weeks.
mom_delta	Month-over-month delta: this month avg – last month avg.
best_day_date	Date of the highest daily_score in the month. The anchor moment.
best_week_start	Start of the highest-scoring week in the month.
biggest_pattern	TEXT — the one correlation insight that needed the full month. Stored as human-readable sentence.
behaviour_shifts	JSONB — directional changes: {snooze_trend: 'down', morning_free_trend: 'up', contentment_trend: 'flat'}

Section 8 — Complete Table Reference

Table	PK	Rows per user	What it stores	Used by
users	user_id	1	Identity, baseline, subscription, personal best	Everything
onboarding_baseline	user_id	1	Calibration answers, synthetic Me Score, pillar weights	Initial dashboard, cold start
user_domains	user_id + domain	1–5	Active pillars and user-defined weights	Me Score, UI routing
alarm_events	alarm_id	1–2/day	Wake and sleep alarms, snooze count, actual wake time	Health section, snooze pattern
morning_screen_events	event_id	1/day	Phone usage in first 30 min after waking	Health report, focus correlation

Table	PK	Rows per user	What it stores	Used by
morning_stamps	stamp_id	1/day	Day type, energy, mood, today's intention	Intent accuracy, energy arc
app_open_intents	intent_id	Multiple/day	Why + how long per guarded app open	Screen section, mindless ratio
focus_sessions	session_id	Multiple/day	Deep work sessions with quality and execution ratio	Career domain, goal progress
midday_energy_checks	check_id	0–1/day	Optional midday energy tap	Energy arc, intraday prediction
evening_checkins	checkin_id	1/day	Day summary, productiveness + contentment 1–5, blockers	Most important daily signal
phone_free_intervals	interval_id	Multiple/day	Phone-idle periods ≥30 min with user activity labels	Weekly phone-free report
sleep_events	sleep_id	1/night	Bedtime, post-declaration screen use, sleep latency	Weekly sleep section
goals	goal_id	Multiple	User goals with deadlines, velocity, progress	Career report, engagement score
subtasks	subtask_id	Multiple/goal	AI-decomposed subtasks with weights and time blocks	Progress bar, session tagging
passive_health	user_id + date	1/day	Sleep, steps, HRV from Health APIs	Health index, sleep correlation
passive_screen	user_id+app+date	Multiple/day	Per-app usage vs intention — Android only	Screen domain score
user_screen_intentions	user_id + app	Multiple	User-set daily limits per guarded app	Adherence calculation
daily_scores	user_id + date	1/day	Nightly pre-computed aggregates. All ML reads this.	All algorithms, weekly report
weekly_reports	user_id + week_start	1/week	Generated Sunday report and Me Score	Report display, MoM comparison

Table	PK	Rows per user	What it stores	Used by
monthly_reports	user_id + month_start	1/month	Monthly arc, MoM delta, biggest pattern, personal bests	Monthly report display
user_baselines	user_id + metric	1/metric	EWMA-smoothed baselines updated nightly	CUSUM, gap calculations, Me Score
user_correlations	user_id + pair	Multiple	Pearson r values for insight generation	Insight section of weekly report
user_alerts	alert_id	Sparse	CUSUM-triggered drift alerts	Wellbeing section, recovery action
user_operating_manual	user_id	1	K-means best conditions. Day 90+ premium.	Premium unlock, best-day profile
notification_events	event_id	Multiple	Notification sends and responses for Thompson Sampling	Bandit training, timing optimisation
commitments	commitment_id	Multiple	'I will try this' taps from recovery actions	Follow-through as product metric

Section 9 — V2 Features (Scoped, Not Built in V1)

These are designed and scoped. Not built in V1. They build on the data foundation V1 creates.

AI Companion Character

A named character the user chooses (strict mentor, gentle friend, honest partner, encouraging parent). In V2 this has three layers: notification tone (same data, different language), weekly report narration (report reads as if that character is speaking), and a conversational interface where the user can ask their character about their week.

Implementation: the companion is a Claude API call with a system prompt defining the character's voice, fed the user's weekly_reports and daily_scores data. The character does not make up insights — it narrates what the data already says, in its own voice.

App Blocking — Hard Intervention

Android: AccessibilityService overlay that triggers when a guarded app is opened past the user's limit. User sees a conscious unlock screen: 'You set a 30-minute limit for YouTube. You have used 28 minutes. Are you choosing to continue?' Yes unlocks for 15 more minutes. No closes the app.

iOS: Screen Time API integration for basic limits. VPN profile trick for harder blocking (adds friction but not impossible). Premium only — user has explicitly opted into being blocked.

Wealth Tracking

Opt-in premium domain. Tracks impulse purchases, savings behaviour, and weekly spend patterns. Only activated if user explicitly wants it. Completely separate pillar with its own data tables. Correlates spend events with stress signals (low contentment days, high screen days).

Me vs ME — Complete Product & Technical Blueprint

Internal document — Version 1.0